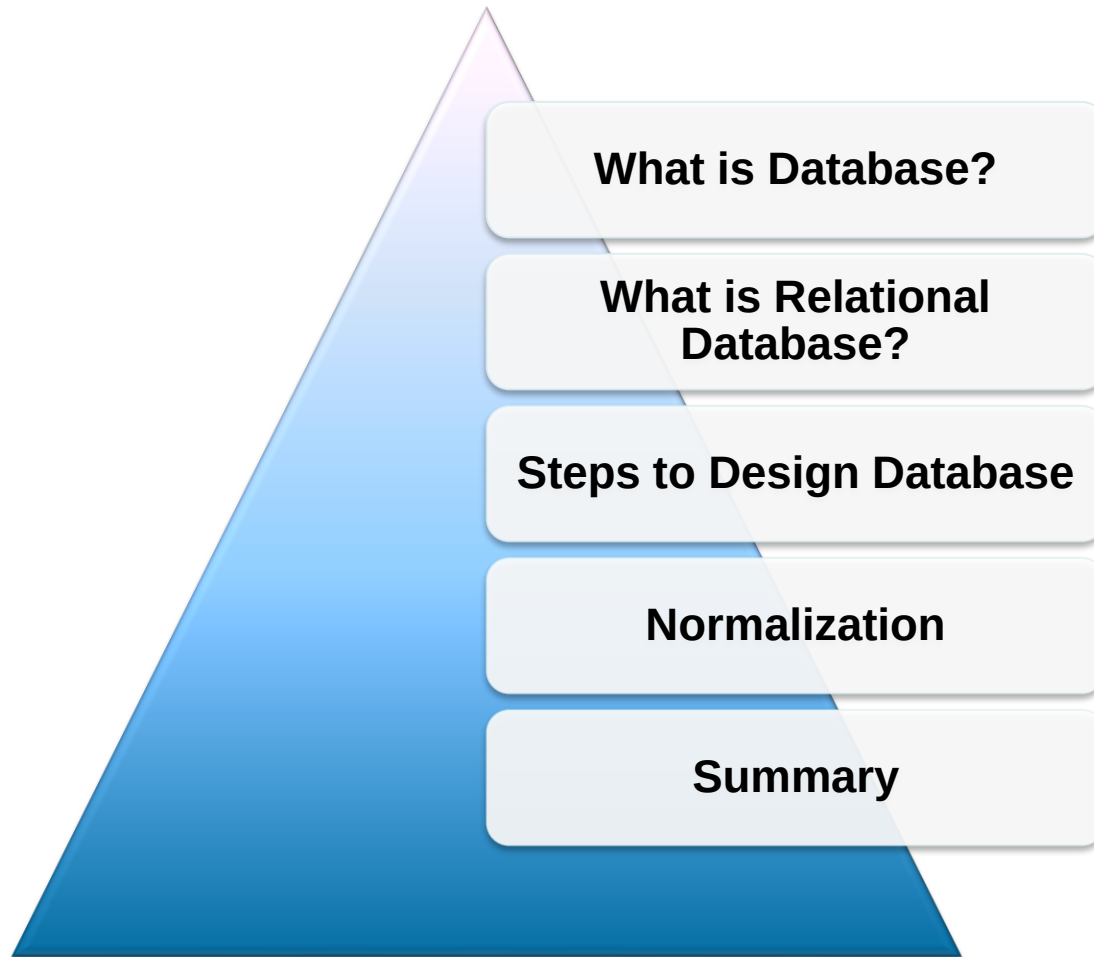




IVY Professional School

Workshop: Relational Database Mgmt & SQL

Agenda





DATABASE BASICS

Introduction

- Database – collection of persistent data
- Database Management System (DBMS) – software system that supports creation, population, and querying of a database

Relational Database

- Relational Database Management System (RDBMS)

- Consists of a number of *tables* and single *schema* (definition of tables and attributes)
- Students (sid, name, login, age, gpa)

Students identifies the table

sid, name, login, age, gpa identify attributes

sid is primary key

An Example Table

- Students (sid: integer, *name*: string, *login*: string, *age*: integer, *gpa*: real)

<u>sid</u>	name	login	age	gpa
50000	Dave	dave@cs	19	3.3
53666	Jones	jones@cs	18	3.4
53688	Smith	smith@ee	18	3.2
53650	Smith	smith@math	19	3.8
53831	Madayan	madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0

Another example: Courses

- Courses (cid, instructor, quarter, dept)

<u>cid</u>	instructor	quarter	dept
Carnatic101	Jane	Fall 06	Music
Reggae203	Bob	Summer 06	Music
Topology101	Mary	Spring 06	Math
History105	Alice	Fall 06	History

One More Example

	number	name	department	salary	location
Row {	23603	Jones	413	1100	New Jersey
	24568	Kerwin	413	2000	New Jersey
	34589	Larson	642	1800	Los Angeles
	35761	Myers	611	1400	Orlando
	47132	Neumann	413	9000	New Jersey
	78321	Stephens	611	8500	Orlando

Primary key

Column

Relational database structure of an Employee table.

Relational Database Model

department	location
413	New Jersey
611	Orlando
642	Los Angeles

Table formed by selecting department and location data from the Employee table.

Keys

- Primary key – minimal subset of fields that is unique identifier for a tuple
 - sid is primary key for Students
 - cid is primary key for Courses
- Foreign key –connections between tables
 - Courses (cid, instructor, quarter, dept)
 - Students (sid, name, login, age, gpa)
 - How do we express which students take each course?

Primary Key Example

<u>State</u>	<u>County</u>	<u>City</u>	Mayor
TX	Bravos	Washington	TTT
TX	Hariss	Washington	FFF
WA	Bravos	Washington	RRR
CA	Hariss	Washington	AAA

**Primary Key with more
than one column is
called as **Composite Key****

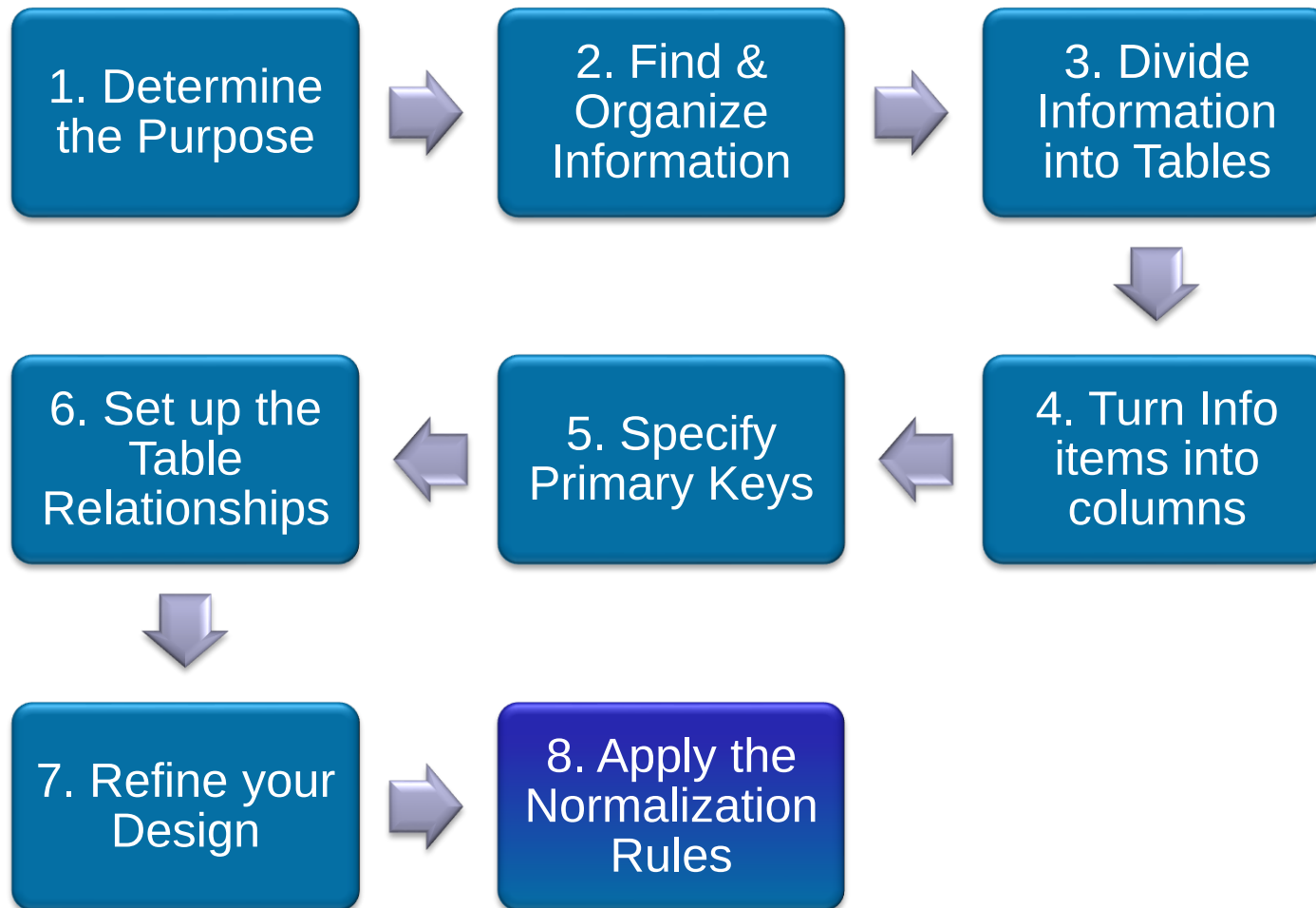


DATABASE DESIGN

What is Good Database Design?

- Divides your information into subject-based tables to reduce redundant data.
- Provides dbms with the information it requires to join the information in the tables together as needed.
- Helps support and ensure the accuracy and integrity of your information.
- Accommodates your data processing and reporting needs.

Database Design Process



1. Determine the Purpose

- How you expect to use it, and who will use it.
- For ex. –
 - The customer database keeps a list of customer information for the purpose of producing mailings and reports.

2. Find & Organize Information

- Start with the existing information
- Next, consider the types of reports you'd need
- Key point – break each piece of info into its smallest useful parts
- Think about the questions you want the database to answer

3. Divide Info into Tables

- Choose the major entities, or subjects.



Customers	Products
Name	Product Name
Address	Price
City, State, Zip	Units in Stock
Send E-mail	Units on Order
Salutation	
E-mail Address	
Suppliers	Orders
Company Name	Order Number
Contact Name	Salesperson
Address	Order Date
City, State, Zip	Product
	Quantity
	Price
	Total

4. Turn Info into Columns

- Determine what information you want to track
- Don't include calculated data
- Store information in its smallest logical parts

Customers	Products
Name	Product Name
Address	Unit Price
City	Units in Stock
Region	Units on Order
Postal Code	Quantity per Unit
Country	
Send E-mail	Orders
Salutation	Order Number
E-mail address	Salesperson
	Order Date
Suppliers	Product
Company Name	Quantity
Contact Name	Price
Address	
City	
Region	
Postal Code	
Country	
Phone	

5. Specifying Primary Keys

- Might be a good idea to have an AutoNumber data type 'ID' field as primary key
- Two or more columns -> Composite Key

Customers CustomerID Name Address City Region Postal Code Country Send E-mail Salutation E-mail address	Products ProductID Product Name Unit Price Units in Stock Units on Order Quantity per Unit
Suppliers SupplierID Company Name Contact Name Address City Region Postal Code Country Phone	Orders OrderID Salesperson Order Date Product Quantity Price

6. Creating Relationships

Orders

1 **Bill To:** Alfreds Futterkiste
Obere Str. 57
Berlin 12209
Germany

2 **Ship To:** Alfreds Futterkiste
Obere Str. 57
Berlin 12209
Germany

3 **Salesperson:** Suyama, Michael

4 **Order ID:** 10643 **Order Date:** 25-Aug-1997 **Required Date:** 22-Sep-1997

5 **Ship Via:** ☒ Speedy ☐ United ☐ Federal

Products:	Unit Price:	Quantity:	Discount:	Extended Price:
Spegesild	\$12.00	2	25%	\$18.00
Chartreuse verte	\$18.00	21	25%	\$283.50
Rossle Sauerkraut	\$45.60	15	25%	\$513.00
*			0%	

Print Invoice

Subtotal: \$814.50
Freight: \$29.46
Total: \$843.96

1. Info from Customers Table
2. Employee Table
3. Orders Table
4. Products Table
5. Order Details Table

6. Creating Relationship contd..

- One to many (Parent – child)
 - Primary Key from Parent Table becomes Foreign Key in Child Table
- Many to Many
- One to One

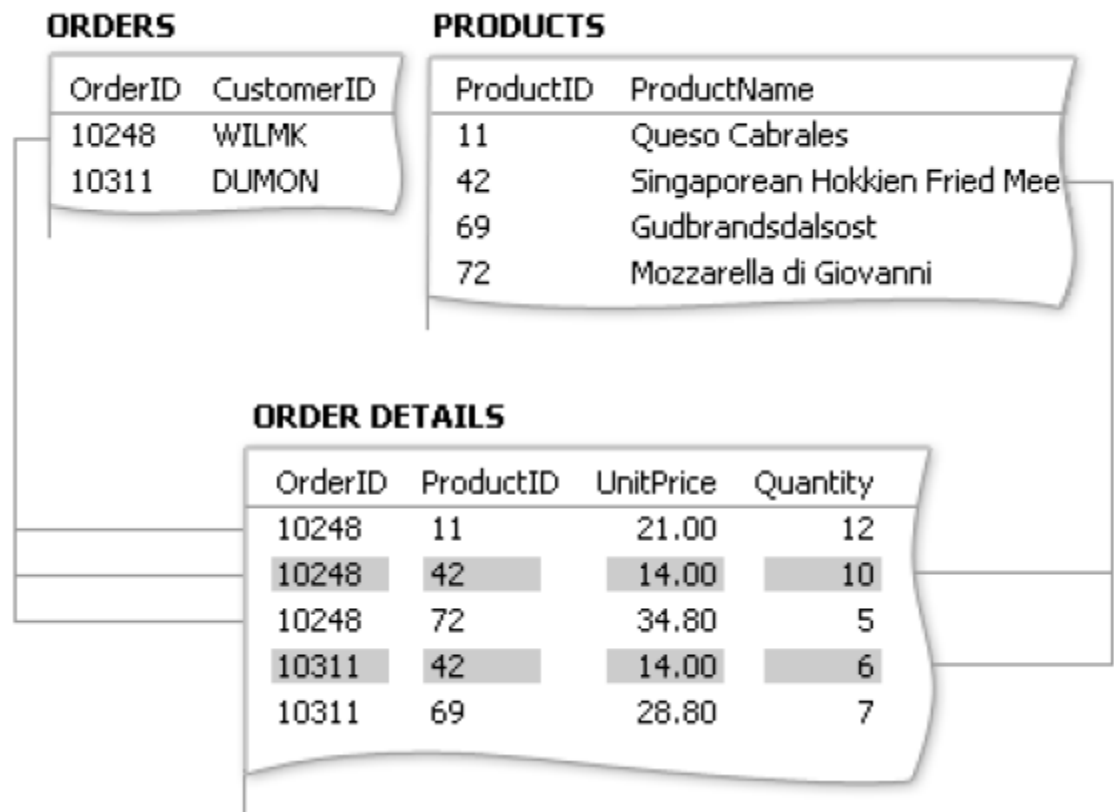
One to Many Relationship

- The supplier ID column in the Products table is called a foreign key

Customers CustomerID Name Address City Region Postal Code Country Send E-mail Salutation E-mail address	Products ProductID Product Name Unit Price Units in Stock Units on Order Quantity per Unit SupplierID
Suppliers SupplierID Company Name Contact Name Address City Region Postal Code Country Phone	Orders OrderID Salesperson Order Date Product Quantity Price

Many to Many Relationship

- Create a third table – Junction Table



7. Refine Design

- Discover room for improvement-
 - Did you forget any columns?
 - Are there un-necessary calculated columns?
 - Are you repeatedly entering duplicate information?
 - Do you have tables with many fields, a limited number of records, & many empty fields in individual records?

8. Apply the Normalization

- Three Most Frequently Use Normal Forms
 - **First Normal Form** -
 - Each row is unique or each table should've a primary key
 - Each cell carries a single value, & never a list of values
 - **Second Normal Form** – Meets 1NF and
 - Each non-primary key column be fully dependent on the entire primary key, and not just parts.
 - **Third Normal Form** – Meets 2NF and
 - Non-key columns should be independent of each other

Test question 1

**What is the function of a primary key?
(Pick one answer.)**

1. To uniquely identify each record in a table.
2. To encrypt and decrypt your database.
3. To help ensure you enter data in the correct table.

Test question 1: Answer

To uniquely identify each record in a table.

All your tables must have a primary key field.

Test question 2

A good database design helps ensure that your data is what? (Pick one answer.)

1. Always backed up.
2. Complete and accurate.
3. Duplicated so it's easier to find.

Test question 2: Answer

Complete and accurate.

Completeness and accuracy are essential for making sound decisions.

Test question 3

How many tables should a well-designed database contain? (Pick one answer.)

1. As many as necessary to capture all your data without redundancy.
2. One, with many fields.
3. Two.

Test question 3: Answer

As many as necessary to capture all your data without redundancy.

That can be one table, or dozens.

DATABASE DESIGN EXERCISES

1. Books Database

- An online book review website is designing their database to track books. They track their books by authors and their titles.
- Please help design this database with the requisite relationships.

2. Cakes & Fudge Sales (H/W)

- Design a database for Cakes & Fudge, a mail-order company that serves chocolate delicacies to a large audience of chocolate-crazed customers. Their daring chefs are always innovating, and they need a better way to manage their ever-growing catalog of chocolate goodness. They also need a way to keep track of customers and the orders they make.

3. The Music School (H/W)

- Cacophoné Studios runs a medium-sized music school. They have a fixed series of courses in mind, and a roster of teachers that can fill in for most of them. They also have a long list of past and potential customers. Last year, a small catastrophe happened when 273 students were crammed into the same class and no teacher was assigned to teach it. (Next door, a class of 14 had somehow ended up with three instructors.) They're hoping that Access can help them avoid the same embarrassment this time around.

Relational Database Overview

Field	Description
authorID	Author's ID number in the database. In the Books.mdb database, this Integer column is defined as auto-increment. For each new row inserted in this table, the database increments the authorID value, ensuring that each row has a unique authorID . This column represents the table's primary key.
firstName	Author's first name (a String).
lastName	Author's last name (a String).
Authors table from Books.mdb .	

Relational Database Overview

authorID	firstName	lastName
1	Harvey	Deitel
2	Paul	Deitel
3	Tem	Nieto
4	Kate	Steinbuhler
5	Sean	Santry
6	Ted	Lin
7	Praveen	Sadhu
8	David	McPhie
9	Cheryl	Yaeger
10	Marina	Zlatkina
11	Ben	Wiedermann
12	Jonathan	Liperi
Data from the Authors table of Books.mdb.		

Relational Database Overview

Field	Description
publ i sherID	The publisher's ID number in the database. This auto-incremented Integer is the table's primary key.
publ i sherName	The name of the publisher (a String).
Publ i shers table from Books.mdb.	

publ i sherID	publ i sherName
1	Prentice Hall
2	Prentice Hall PTG
Data from the Publ i shers table of Books.mdb.	

Field	Description
authorID	The author's ID number, which allows the database to associate each book with a specific author. The integer ID number in this column must also appear in the Authors table.
i s bn	The ISBN number for a book (a String).
AuthorISBN table from Books.mdb.	

Relational Database Overview

authorID	isbn	authorID	isbn
1	0130895725	2	0139163050
1	0132261197	2	013028419x
1	0130895717	2	0130161438
1	0135289106	2	0130856118
1	0139163050	2	0130125075
1	013028419x	2	0138993947
1	0130161438	2	0130852473
1	0130856118	2	0130829277
1	0130125075	2	0134569555
1	0138993947	2	0130829293
1	0130852473	2	0130284173
1	0130829277	2	0130284181
1	0134569555	2	0130895601
1	0130829293	3	013028419x
1	0130284173	3	0130161438
1	0130284181	3	0130856118
1	0130895601	3	0134569555
2	0130895725	3	0130829293
2	0132261197	3	0130284173
2	0130895717	3	0130284181
2	0135289106	4	0130895601

Data from AuthorISBN table in Books .mdb.

Relational Database Overview

Field	Description
isbn	ISBN number of the book (a String).
title	Title of the book (a String).
editionNumber	Edition number of the book (a String).
copyright	Copyright year of the book (an Integer).
description	Description of the book (a String).
publisherID	Publisher's ID number (an Integer). This value must correspond to an ID number in the Publishers table.
imageFile	Name of the file containing the book's cover image (a String).
price	Suggested retail price of the book (a real number). [Note: The prices shown in this database are for example purposes only.]
Titles table from Books.mdb.	

Relational Database Overview

isbn	title	edition-Number	publisherID	copy-right	price
0130923613	Python How to Program	1	1	2002	\$69.95
0130622214	C# How to Program	1	1	2002	\$69.95
0130341517	Java How to Program	4	1	2002	\$69.95
0130649341	The Complete Java Training Course	4	2	2002	\$109.95
0130895601	Advanced Java 2 Platform How to Program	1	1	2002	\$69.95
0130308978	Internet and World Wide Web How to Program	2	1	2002	\$69.95
0130293636	Visual Basic .NET How to Program	2	1	2002	\$69.95
0130895636	The Complete C++ Training Course	3	2	2001	\$109.95
0130895512	The Complete e-Business & e-Commerce Programming Training Course	1	2	2001	\$109.95
Portion of the data from the <code>Titles</code> table of <code>Books.mdb</code> .					

Relational Database Overview

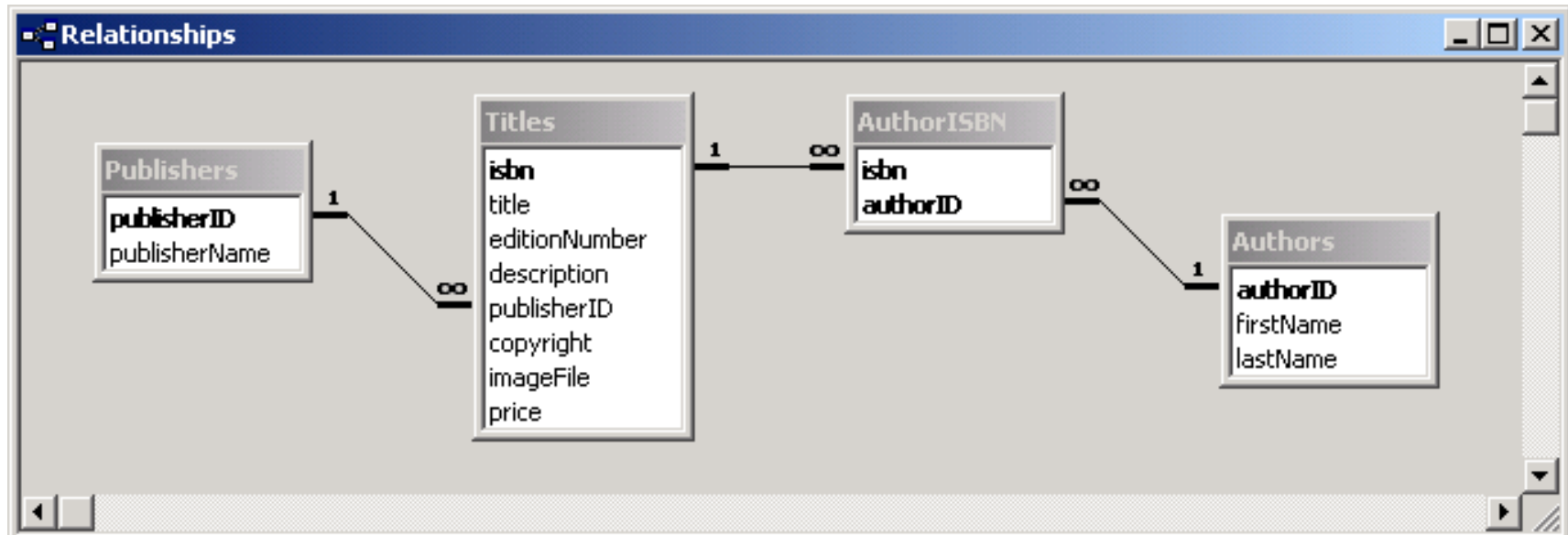


Table relationships in Books . mdb.

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